

NE-B Failure as Hilbert–Pólya Detection: Universality of the v2.0 Weil Quadratic Form on the Selberg Zeta Function

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Abstract

The v2.0 methodology recently proposed for the Riemann Hypothesis (Geiger 2026, Part II) reduces RH to *even dominance* of the Weil quadratic form QW_λ via three algebraic and analytic ingredients: the Shift Parity Lemma, frontier-prime dominance, and two non-existence theorems (NE-A: non-positivity of the prime shift multiplier $M(\xi) = -2 \operatorname{Re}[\zeta' / \zeta(\frac{1}{2} + i\xi)]$, NE-B: no universal commuting operator). A natural question is whether this framework is specific to the Riemann zeta function or whether it is a general-purpose method for explicit-formula zeta structures. We test it on the Selberg zeta function $Z_\Gamma(s)$ of a compact hyperbolic surface $\Gamma \backslash \mathbb{H}$. We show:

- (i) The Shift Parity Lemma transfers unchanged (it is purely algebraic).
- (ii) Frontier dominance transfers with modified constants, accounting for the exponential density of primitive closed geodesics $\pi_\Gamma(T) \sim e^T/T$ in place of $\pi(x) \sim x/\log x$.
- (iii) An analogue QW_λ^Γ of the Weil quadratic form is well defined via the geometric side of the Selberg trace formula.
- (iv) *Even dominance of QW_λ^Γ* furnishes a second, independent proof of the Selberg Riemann Hypothesis, redundant in view of Selberg’s spectral proof via the Laplacian Δ_Γ .
- (v) *NE-B fails in the Selberg setting.* The Laplacian Δ_Γ on $\Gamma \backslash \mathbb{H}$ commutes with all geodesic shift operators and hence is a universal commutant. The Hilbert–Pólya program is realized here by Δ_Γ itself.

The failure of NE-B in the Selberg setting is not a defect; it reflects the geometric coordination of geodesics on a common surface, which is absent for the primes. We deduce a meta-principle: the v2.0 framework is *universal*, while Hilbert–Pólya is a *special case* whose existence depends on whether the underlying counting objects admit a hidden common symmetry. In the Selberg case they do; in the Riemann case they do not (by NE-B). The Weil quadratic form, not the spectral operator, is the fundamental object.

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1 Introduction

1.1 Motivation and main claim

The v2.0 programme for the Riemann Hypothesis [11] reduces RH to a quadratic-form inequality. Specifically, Connes’ 2026 survey [8] writes RH as the statement that the minimum eigenvalue of a one-parameter family of quadratic forms QW_λ on $L^2([-\log \lambda, \log \lambda])$ is simple and has an even eigenfunction. The v2.0 approach establishes this in three layers:

- a purely algebraic *Shift Parity Lemma* (Theorem 4.1 of [11]), showing that the even–odd difference matrix $D_N(r)$ has strictly negative minimum eigenvalue for all $r \in (0, 2)$, $N \geq 3$;
- an analytic *frontier-dominance* mechanism: the gap $\Delta(\lambda) = \lambda_1^+ - \lambda_1^-$ is driven by primes with $\log p / \log \lambda$ close to 1 (“frontier primes”), whose density is controlled by the Prime Number Theorem;
- two structural non-existence results that close off alternative routes: *NE-A* (the prime shift multiplier $M(\xi)$ is not nonnegative a.e., ruling out PF_∞ total positivity) and *NE-B* (no nontrivial symmetric matrix commutes with all $D_N(r)$ simultaneously, ruling out Sturm–Liouville-type universal operators at the tested truncations).

The v2.0 programme succeeds *without* constructing a Hilbert–Pólya operator. It is natural to ask whether this methodological package transfers to other zeta-like functions, and, if so, what it teaches us. The paradigmatic case is the Selberg zeta function $Z_\Gamma(s)$ of a compact hyperbolic surface $M = \Gamma \backslash \mathbb{H}$, where:

- the geometric explicit formula (Selberg trace formula [17]) plays the role of the Weil explicit formula [4];
- the lengths of primitive closed geodesics play the role of primes;
- the Selberg Riemann Hypothesis (all non-trivial zeros on the critical line $\text{Re}(s) = 1/2$) is a theorem, via the self-adjointness of the Laplacian Δ_Γ .

Thus we ask two concrete questions:

Q1. *Do the v2.0 ingredients (Shift Parity Lemma, frontier dominance, Weil quadratic form, NE-A, NE-B) all transfer to the Selberg setting?*

Q2. *If any of them fails to transfer, what is the structural reason?*

Our answers, in brief:

- Shift Parity Lemma: **transfers unchanged** (algebraic).
- Frontier dominance: **transfers with modified constants**, reflecting $\pi_\Gamma(T) \sim e^T/T$.
- Weil quadratic form: **natural analogue** QW_λ^Γ via the geometric side of the trace formula.

- NE-A: **transfers** – the Selberg analogue multiplier is also oscillatory (each non-trivial zero of Z_Γ contributes a negative excursion).
- NE-B: **fails**. The Laplacian Δ_Γ commutes with every geodesic translation operator: a universal commutant exists, by the very definition of the hyperbolic surface geometry.

The failure of NE-B is the scientifically interesting outcome. It is not a failure of v2.0; it is a *diagnostic*. Where NE-B fails, the Hilbert–Pólya programme succeeds (the operator Δ_Γ is a ready-made commutant). Where NE-B holds (as we have argued, up to the full-space conjecture, for the Riemann zeta), no universal commuting operator exists and Hilbert–Pólya in the strict sense cannot be realized. In either case the v2.0 framework applies: it is the common structural substrate.

Remark 1.1 (Position within the meta-framework). This paper operates as the *Lie-origin companion* to the arithmetic Riemann case [11] and sits within the three-component decomposition $\text{RH}_X = \text{GBC}_X + C2_X + \text{Hurwitz}_X$ developed in the companion meta-paper [12]. Selberg is the paradigmatic case in which both substeps of $C2_X$ are trivialised by a Lie-theoretic universal commutant; Riemann is the case in which the parity substep is closed by v2.1 but the eigenvector-realisation substep remains open. See Theorem 6.2 below for details.

1.2 Main theorem (informal)

Theorem 1.2 (v2.0 Selberg transfer, informal). *Let $M = \Gamma \backslash \mathbb{H}$ be a compact hyperbolic surface and $Z_\Gamma(s)$ its Selberg zeta function. The v2.0 methodology applies to the Selberg quadratic form QW_λ^Γ and yields a proof of the Selberg Riemann Hypothesis by even dominance. The analogue of NE-A holds, but NE-B fails: the Laplace–Beltrami operator Δ_Γ is a universal commutant for the family of geodesic shift operators. The resulting proof of Selberg-RH is therefore redundant (a second proof), and this redundancy exposes a meta-principle: the Weil quadratic form is universal, whereas a Hilbert–Pólya operator exists if and only if NE-B fails.*

1.3 Structure of the paper

Section 2 recalls the v2.0 ingredients. Section 3 sets up the Selberg trace formula and the geodesic analogue of the explicit formula. Section 4 carries out the transfer of the Shift Parity Lemma, frontier dominance, and constructs QW_λ^Γ ; a second proof of Selberg-RH is derived. Section 5 proves the NE-B failure and constructs the universal commutant Δ_Γ explicitly in the transfer-operator formalism. Section 6 draws the meta-principle: Hilbert–Pólya is a special case of v2.0. Section 7 discusses applications and outlook (generalised L-functions, Ruelle zeta, quantum chaos). Section 9 concludes.

2 Recollection of the v2.0 Method

We briefly summarize the four key ingredients of the v2.0 approach to RH. A full treatment is in [11].

2.1 The Weil quadratic form

Set $L = \log \lambda$. On $L^2([-L, L])$, define

$$\langle A_\lambda f \mid f \rangle = (\log 4\pi + \gamma) \|f\|^2 + \int_0^\infty \left[\langle T_u f + T_{-u} f, f \rangle - 2e^{-u/2} \|f\|^2 \right] \frac{e^{u/2}}{2 \sinh u} du + \sum_p \sum_{m \geq 1} \frac{\log p}{p^{m/2}} \langle T_{m \log p} f + T_{-m \log p} f, f \rangle \quad (1)$$

where T_s is the shift operator $(T_s f)(t) = f(t-s) \mathbf{1}_{[-L, L]}(t-s)$ and γ is the Euler–Mascheroni constant. The quadratic form QW_λ is the quadratic form of A_λ . The key algebraic tool is Theorem 6.1 of [8] (proved jointly with van Suijlekom, building on [9]):

Theorem 2.1 ([8, Theorem 6.1]). *Let $L > 0$, let $\mathcal{D}$ be a real distribution on the interval $[0, L]$, and let $\tilde{\mathcal{D}}$ be the associated even distribution on $[-L, L]$. Assume that the quadratic form with Schwartz kernel $\tilde{\mathcal{D}}(x-y)$ defines a lower-bounded selfadjoint operator on $L^2([-L/2, L/2])$, and that the minimum of its spectrum is a simple, isolated eigenvalue with even eigenfunction η . Then all zeros of the entire function $\tilde{\eta}(z)$, $z \in \mathbb{C}$ (Fourier transform of η), lie on the real line.*

Applied to the Weil quadratic form QW_λ (with $\mathcal{D}$ the Weil distribution from the prime explicit formula), if the minimum eigenvalue is simple with even eigenfunction for all $\lambda \geq \lambda_0$ and the corresponding eigenvectors converge to Riemann’s Ξ -function, then the zeros of $\tilde{\eta}$ coincide with those of Ξ ; hence RH holds. The convergence step remains open [8, §6.6].

Remark 2.2 (Interval convention). Theorem 2.1 places the quadratic-form operator on $L^2([-L/2, L/2])$ in Connes’ notation. In the present paper (Section 4 onwards), test functions live on $L^2([-L, L])$. Applying Theorem 2.1 requires the identification $L_{\text{Thm}} = 2L$, i.e. the half-length in Connes’ statement equals the full truncation length L .

Remark 2.3 (Independence from Connes’ convergence step). Theorem 2.1 is drawn from Connes’ 2026 preprint [8]; the convergence of the construction to the Weil distribution is noted as still open [8, §6.6]. In the Selberg application (Section 4), this open step is *immaterial*: the conclusion (Selberg-RH) is independently secured by Selberg’s classical spectral proof via the Laplacian Δ_Γ [17]. Theorem 2.1 plays a structural consistency role, not a logically necessary one. The broader programme of encoding L -functions via spectral quadratic forms is additionally supported by Connes’ 1999 noncommutative geometry approach [7], Deninger’s regularized-determinant programme [2], and the Bost–Connes thermodynamical system [1].

2.2 The Shift Parity Lemma

Let $\varphi_n(t) = \cos(n\pi t/L)/\sqrt{L}$ (even basis) and $\psi_n(t) = \sin((n+1)\pi t/L)/\sqrt{L}$ (odd basis). Define the *difference matrix* $D_N(r)$ by

$$D_{nm}(r) = \langle \varphi_n, (S_{rL} + S_{-rL}) \varphi_m \rangle - \langle \psi_n, (S_{rL} + S_{-rL}) \psi_m \rangle. \quad (2)$$

Theorem 2.4 (Shift Parity Lemma, [11]). *For every $N \geq 3$ and every $r \in (0, 2)$, $\lambda_{\min}(D_N(r)) < 0$.*

This is a purely algebraic, closed-form statement about trigonometric matrices; the entries of $D_N(r)$ are explicit sums of sin and cos with linear and constant coefficients in r .

2.3 Frontier dominance

The cumulative even–odd gap $\Delta(\lambda) = \lambda_1^+(\lambda) - \lambda_1^-(\lambda)$ is driven by primes in the *frontier band* $[\lambda/e, \lambda]$: their count is $\pi(\lambda) - \pi(\lambda/e) \sim \lambda/\log \lambda$ by the PNT, and they are the ones for which $\log p/L \in [1 - 1/\log \lambda, 1]$. By Theorem 2.4, each such prime contributes a negative summand of order $-c \log p/\sqrt{p}$ to the difference, with the sum dominating any archimedean or small-prime correction. Numerically, in [11] frontier primes are shown to contribute $> 99\%$ of the gap for all tested $\lambda \leq 1.3 \cdot 10^6$.

2.4 NE-A and NE-B

Define the *prime shift multiplier*

$$M(\xi) = -2 \operatorname{Re} \left[\frac{\zeta'}{\zeta} \left(\frac{1}{2} + i\xi \right) \right]. \quad (3)$$

Theorem 2.5 (NE-A, [11]). $\{\xi \in \mathbb{R} : M(\xi) < 0\}$ has positive Lebesgue measure. In particular, the prime shift operator is not PF_∞ -total positive.

Theorem 2.6 (NE-B, [11]). For $N \in \{3, 5, 7, 10, 15\}$ and any sufficiently dense sample of $r \in (0, 2)$, the only symmetric matrix $T \in \operatorname{Sym}_N(\mathbb{R})$ satisfying $[T, D_N(r_i)] = 0$ for all i is $T = cI$. Consequently no nontrivial universal commuting operator exists at these truncations; the full-space conjecture [11, Conjecture 5.3] asserts the same for all N .

NE-B is the technically decisive structural result: it expresses that the primes have no hidden common symmetry. The conceptual content is multiplicative independence.

3 The Selberg Setting

We briefly record the standard setup.

3.1 Compact hyperbolic surfaces

Let $M = \Gamma \backslash \mathbb{H}$, where $\mathbb{H} = \{x + iy : y > 0\}$ is the upper half-plane with metric $ds^2 = (dx^2 + dy^2)/y^2$ and $\Gamma \subset \operatorname{PSL}(2, \mathbb{R})$ is a cocompact discrete torsion-free subgroup. Then M is a compact hyperbolic surface of genus $g \geq 2$. The Laplace–Beltrami operator Δ_Γ acts on $L^2(M)$ as a non-negative self-adjoint operator with purely discrete spectrum

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots, \quad \lambda_j \rightarrow \infty. \quad (4)$$

Writing $\lambda_j = 1/4 + r_j^2$ with $r_j \in \mathbb{R} \cup i[-1/2, 1/2]$, the spectral parameters r_j encode the quantum spectrum.

3.2 Primitive closed geodesics

The conjugacy classes of hyperbolic elements in Γ correspond bijectively to (unoriented) closed geodesics on M . A geodesic is *primitive* if it is not a positive iterate of another. Denote by $\mathcal{P}_\Gamma$ the set of primitive closed geodesics, and for $\gamma \in \mathcal{P}_\Gamma$ by ℓ_γ its length.

The Prime Geodesic Theorem [14, 16] states

$$\pi_\Gamma(T) := \#\{\gamma \in \mathcal{P}_\Gamma : \ell_\gamma \leq T\} \sim \frac{e^T}{T}, \quad T \rightarrow \infty. \quad (5)$$

This is the geodesic analogue of the PNT. Crucially, *the density is exponential, not polynomial*: $\pi_\Gamma(T)$ grows like e^T/T , whereas $\pi(x)$ grows like $x/\log x = (e^{\log x})/\log x$. In other words, the “prime-length-spectrum” of geodesics, seen on the log-axis, has density e^T/T (in the variable $T = \ell_\gamma$), comparable to the density of primes on the log-axis (in the variable $u = \log p$) only after the change of variable.

3.3 Selberg zeta function

Definition 3.1. The Selberg zeta function is

$$Z_\Gamma(s) = \prod_{\gamma \in \mathcal{P}_\Gamma} \prod_{k=0}^{\infty} (1 - e^{-(s+k)\ell_\gamma}), \quad \operatorname{Re}(s) > 1. \quad (6)$$

$Z_\Gamma(s)$ extends meromorphically to $\mathbb{C}$, with:

- *Trivial zeros* at $s = -k$, $k \in \mathbb{N}_0$ (and further trivial zeros associated with topology; see [13]).
- *Non-trivial zeros* at $s = 1/2 \pm ir_j$ for each Laplace eigenvalue $\lambda_j = 1/4 + r_j^2 > 0$.
- A zero at $s = 1$ (from $\lambda_0 = 0$).

Theorem 3.2 (Selberg Riemann Hypothesis, [17]). *All non-trivial zeros of $Z_\Gamma(s)$ lie on the line $\operatorname{Re}(s) = 1/2$.*

This follows from the non-negativity of Δ_Γ : $\lambda_j \geq 0$ implies $r_j \in \mathbb{R} \cup i[-1/2, 1/2]$. Non-negative real r_j (i.e. $\lambda_j \geq 1/4$) give zeros on the critical line $\operatorname{Re}(s) = 1/2$. Exceptional eigenvalues $\lambda_j \in (0, 1/4)$, however, produce $r_j = i\delta_j$ with $\delta_j \in (0, 1/2)$, and thereby yield zeros at $s = 1/2 \pm \delta_j$ that are *not* on the critical line. Selberg-RH in the strict sense (all non-trivial zeros on $\operatorname{Re}(s) = 1/2$) therefore requires the spectral gap condition $\lambda_1(\Delta_\Gamma) \geq 1/4$. This is Selberg’s eigenvalue conjecture [18]: it is established in the weaker form $\lambda_1 \geq 3/16$ for congruence subgroups but remains open as $\lambda_1 \geq 1/4$ in full generality. The v2.0 analysis in Section 4 is carried out under this hypothesis.

Remark 3.3 (Functional equation). The completed Selberg zeta function satisfies the functional equation $Z_\Gamma(s) = Z_\Gamma(1-s) \cdot \Xi_\Gamma(s)$, where $\Xi_\Gamma(s)$ is an explicit meromorphic factor depending on $\operatorname{Area}(M)$ and the genus of M [13]. In particular, non-trivial zeros occur in pairs $\rho, 1-\rho$, confirming their symmetry about $\operatorname{Re}(s) = \frac{1}{2}$.

3.4 Selberg explicit formula

The Selberg trace formula [17, 13] provides the geodesic analogue of the Weil explicit formula. For a pair $(h, \hat{h})$ of sufficiently rapidly decaying functions related by Fourier transform, the trace formula reads

$$\sum_{j=0}^{\infty} h(r_j) = \frac{\operatorname{Area}(M)}{4\pi} \int_{\mathbb{R}} h(r) r \tanh(\pi r) dr + \sum_{\gamma \in \mathcal{P}_\Gamma} \sum_{m=1}^{\infty} \frac{\ell_\gamma \hat{h}(m\ell_\gamma)}{2 \sinh(m\ell_\gamma/2)}. \quad (7)$$

The left-hand side is the *spectral side* (a sum over eigenvalues of Δ_Γ); the right-hand side is the *geometric side* (an identity term plus a sum over geodesics). This plays exactly the role of

$$\sum_{\rho} \Phi(\rho) = (\text{archimedean term}) - \sum_p \sum_{m \geq 1} \frac{\log p}{p^{m/2}} (\hat{\Phi}(m \log p) + \hat{\Phi}(-m \log p))$$

in the Weil explicit formula: primes $\leftrightarrow$ primitive geodesics, $\log p \leftrightarrow \ell_\gamma$, non-trivial zeros $\rho \leftrightarrow$ spectral parameters r_j .

4 Applying v2.0 to Selberg

4.1 Selberg geodesic shifts and Shift Parity Lemma

Set $L = \log \lambda$. On $L^2([-L, L])$, define the Selberg prime shift operator

$$(P_\lambda^\Gamma f)(t) = \sum_{\gamma \in \mathcal{P}_\Gamma, \ell_\gamma \leq 2L} \sum_{m=1}^{\lfloor 2L/\ell_\gamma \rfloor} \frac{\ell_\gamma}{2 \sinh(m\ell_\gamma/2)} [f(t - m\ell_\gamma) + f(t + m\ell_\gamma)] \mathbf{1}_{[-L, L]}. \quad (8)$$

In the cosine / sine basis of Section 2, define the *Selberg difference matrix*

$$D_{nm}^\Gamma(r) = \langle \varphi_n, (S_{rL} + S_{-rL}) \varphi_m \rangle - \langle \psi_n, (S_{rL} + S_{-rL}) \psi_m \rangle. \quad (9)$$

This is *the same matrix* $D_N(r)$ as in the Riemann setting: it depends only on the normalized shift r , not on the source of the shift (prime length or geodesic length). Consequently:

Proposition 4.1 (Transfer of Shift Parity). *The Shift Parity Lemma (Theorem 2.4) holds verbatim in the Selberg setting: for every $N \geq 3$, every $r \in (0, 2)$, $\lambda_{\min}(D_N^\Gamma(r)) < 0$.*

Proof. $D_N^\Gamma(r) = D_N(r)$ as matrices. The Shift Parity Lemma depends only on the harmonic structure of the cosine and sine bases on $[-L, L]$, and on the fact that $S_{rL} + S_{-rL}$ is the sum of two shifts by the same amount in opposite directions. The number-theoretic or geometric origin of r is immaterial. $\square$

Remark 4.2. This is the algebraic universality. In the v2.0 framework for the Riemann zeta, the Shift Parity Lemma is the bedrock on which everything rests. It transfers to *every* setting where zeros are parametrized by an explicit formula with real, positive lengths playing the role of $\log p$. This includes Dirichlet L-functions, Dedekind zeta functions, Selberg zeta, Ruelle zeta, and in principle any Hecke or automorphic L-function.

4.2 Selberg Weil quadratic form

On $L^2([-L, L])$, define the *Selberg–Weil quadratic form*

$$\text{QW}_\lambda^\Gamma(f) = \frac{\text{Area}(M)}{4\pi} K_{\text{arch}}^\Gamma(f, f) + \sum_{\gamma \in \mathcal{P}_\Gamma, \ell_\gamma \leq 2L} \sum_{m=1}^{\infty} \frac{\ell_\gamma}{2 \sinh(m\ell_\gamma/2)} \langle T_{m\ell_\gamma} f + T_{-m\ell_\gamma} f, f \rangle, \quad (10)$$

Remark 4.3 (Truncation is exact). For $f \in L^2([-L, L])$ and $|u| > 2L$, the shift inner product vanishes: $\langle T_u f + T_{-u} f, f \rangle = 0$, because f is supported on $[-L, L]$ while $T_{\pm u} f$ has support shifted outside $[-L, L]$. Consequently every geodesic iterate with $m\ell_\gamma > 2L$ contributes *exactly* zero to QW_λ^Γ , with no approximation error. The sum in (10) is therefore not an approximation but an exact expression for the quadratic form on $L^2([-L, L])$.

where K_{arch}^Γ is the archimedean analogue

$$K_{\text{arch}}^\Gamma(f, f) = \int_0^\infty [\langle T_u f + T_{-u} f, f \rangle - 2 c_\Gamma^{\text{id}}(u) \|f\|^2] \cdot \kappa^\Gamma(u) du, \quad (11)$$

obtained by translating the identity term in (7) into shift operators.

Remark 4.4 (Convergence and renormalisation at $u = 0$). Equation (11) is the u -domain expression of the identity term in the Selberg trace formula applied to $h(r) := |\hat{f}(r)|^2$; for f in a dense subspace (e.g. $C_c^\infty([-L, L])$), both sides converge absolutely. The pointwise singularity of κ^Γ at $u = 0$ is matched by that of c_Γ^{id} , and the bracket is the regularised identity contribution; cf. [13, Ch. 6].

The coefficient

$$c_\Gamma^{\text{id}}(u) := \frac{\text{Area}(M)}{4\pi} \int_0^\infty r \tanh(\pi r) \cos(ru) dr \quad (12)$$

is the Fourier-cosine transform of the Plancherel measure density $r \tanh(\pi r)$, encoding the identity-coset contribution from (7); the integral is interpreted in the tempered-distribution sense since $r \tanh(\pi r)$ is polynomially bounded (cf. [13, Ch. 6]). Here

$$\kappa^\Gamma(u) := \frac{1}{4\pi} \frac{d}{du} \left[\frac{1}{\sinh(u/2)} \right], \quad u > 0, \quad (13)$$

is the Plancherel kernel of the Selberg transform [15, eq. (1.40)]. Note that $\kappa^\Gamma(u) < 0$ for all $u > 0$, since $\frac{d}{du}[\sinh(u/2)^{-1}] = -\frac{\cosh(u/2)}{2\sinh^2(u/2)} < 0$. This is consistent with the positivity structure of K_{arch}^Γ : the bracket in (11) is negative when $c_\Gamma^{\text{id}}(u)$ normalises the identity operator, and the product of two negative quantities is positive.

The quadratic form QW_λ^Γ admits a parity decomposition. Its minimum eigenvalue is attained by either an even or an odd eigenfunction.

Definition 4.5 (Selberg even dominance). We say that *even dominance* holds for QW_λ^Γ at parameter λ if the minimum eigenvalue of QW_λ^Γ is simple and attained by an even eigenfunction.

Remark 4.6 (Trace formula as bridge between spectral and geometric spaces). The form QW_λ^Γ is defined entirely in the spectral parameter space $L^2([-L, L])$, where T_u denotes the truncated shift $f(\cdot + u)$ restricted to $[-L, L]$. A geometric realization via the geodesic flow on $L^2(SM)$ also exists: both descriptions yield the same spectral sum $\sum_j \hat{h}(r_j)$ when evaluated against test functions in the domain of the Selberg transform. The Selberg trace formula (7) serves as the implicit bridge between the two spaces. An explicit intertwiner via the Selberg–Helgason transform on $L^2(SM)$ is deferred to future work.

4.3 Frontier dominance in the exponential density regime

The frontier band for Selberg is the set of geodesics with ℓ_γ/L close to 1, i.e. $\ell_\gamma \in [L-1, L]$. By the Prime Geodesic Theorem (5),

$$\#\{\gamma : \ell_\gamma \in [L-1, L]\} = \pi_\Gamma(L) - \pi_\Gamma(L-1) \sim \frac{e^L}{L}(1 - e^{-1}) = \frac{\lambda}{\log \lambda}(1 - e^{-1}). \quad (14)$$

This is *the same asymptotic order* as the Riemann frontier count $\pi(\lambda) - \pi(\lambda/e)$, up to the constant $(1 - e^{-1}) \approx 0.632$.

The frontier geodesic weight is, for $\ell_\gamma \sim L$,

$$w_\gamma^\Gamma := \frac{\ell_\gamma}{2 \sinh(\ell_\gamma/2)} \sim \frac{L}{e^{L/2}} = \log \lambda \cdot \lambda^{-1/2}. \quad (15)$$

Compare this with the Riemann frontier prime weight $w_p^{\text{Riem}} = \log p/p^{1/2} \sim L \cdot \lambda^{-1/2}$ for $p \sim \lambda$. The weights agree asymptotically in order of magnitude.

Combining (14) and the weight estimate with the Shift Parity Lemma (which ensures that each frontier summand is negative with magnitude bounded away from zero), we obtain the Selberg analogue of the frontier-dominance estimate:

$$\Delta^\Gamma(\lambda) := \lambda_1^+(\text{QW}_\lambda^\Gamma) - \lambda_1^-(\text{QW}_\lambda^\Gamma) \leq -c_\Gamma \sqrt{\lambda} + o(\sqrt{\lambda}), \quad (16)$$

for a positive constant $c_\Gamma > 0$ depending on the surface M (via $\text{Area}(M)$ and the Selberg zeta function near $s = 1$).

Lemma 4.7 (Frontier dominance constant — lower bound). *For every compact hyperbolic surface M there exists $\lambda_0 = \lambda_0(M) > 0$ such that the frontier dominance constant in (16) satisfies*

$$c_\Gamma \geq \frac{1}{2}(1 - e^{-1}) > 0 \quad \text{for all } \lambda \geq \lambda_0. \quad (17)$$

Remark 4.8. The normalised unsigned frontier sum $\lambda^{-1/2} \sum_{\ell_\gamma \in [L-1, L]} w_\gamma^\Gamma \rightarrow (1 - e^{-1}) \approx 0.632$ as $\lambda \rightarrow \infty$. The factor $\frac{1}{2}$ in (17) is conservative: it uses the quantitative lower bound of [11, Lem. A.3], which gives signed magnitude $\geq \frac{1}{2}w_\gamma^\Gamma$ per frontier geodesic. Proposition 4.1 provides the sign ($\lambda_{\min} < 0$) alone; the magnitude bound is the content of the Part-II lemma. Under the sharp bound, $c_\Gamma \rightarrow (1 - e^{-1})$ asymptotically.

Proof sketch. The frontier band $[L - 1, L]$ contains $(1 - e^{-1})e^L/L(1 + o(1)) = (1 - e^{-1})\lambda/\log \lambda(1 + o(1))$ primitive geodesics by (14). Each contributes weight $w_\gamma^\Gamma = L/(2 \sinh(L/2)) \sim \log \lambda \cdot \lambda^{-1/2}$ by (15). By [11, Lem. A.3], each frontier summand is negative with magnitude at least $\frac{1}{2}w_\gamma^\Gamma$ for all large λ . Summing over the frontier band and dividing by $\sqrt{\lambda}$ gives

$$\sum_{\ell_\gamma \in [L-1, L]} w_\gamma^\Gamma \geq (1 - e^{-1}) \frac{\lambda}{\log \lambda} \cdot \frac{\log \lambda}{\sqrt{\lambda}} \cdot \frac{1}{2}(1 + o(1)) = \frac{1}{2}(1 - e^{-1})\sqrt{\lambda}(1 + o(1)),$$

yielding $c_\Gamma \geq (1 - e^{-1})/2$. □

Remark 4.9. The factor $(1 - e^{-1}) \approx 0.632$ is the geometric ratio of frontier geodesic density (exponential regime) to frontier prime density (polynomial regime): Selberg has $(1 - e^{-1})\lambda/\log \lambda$ frontier geodesics where Riemann has $\lambda/\log \lambda$ frontier primes. This accounts for the only structural difference between the two frontier estimates.

4.4 Conditional reduction via v2.0

Theorem 4.10 (v2.0 Selberg-RH, conditional). *Let $M = \Gamma \backslash \mathbb{H}$ be a compact hyperbolic surface satisfying $\lambda_1(\Delta_\Gamma) \geq 1/4$. Assume even dominance of QW_λ^Γ (Definition 4.5) holds for all $\lambda \geq \lambda_0(\Gamma)$. Then all non-trivial zeros of $Z_\Gamma(s)$ lie on $\text{Re}(s) = 1/2$.*

Proof sketch. The structure of the argument is identical to the v2.0 proof for Riemann.

Step 1 (pointwise even preference, including iterates). By Proposition 4.1, for every $\gamma \in \mathcal{P}_\Gamma$ and every $m \geq 1$ with $m\ell_\gamma \leq 2L$, the difference matrix $D_N^\Gamma(m\ell_\gamma/L)$ has $\lambda_{\min} < 0$. Here $m > 1$ corresponds to the m -fold iterate of γ ; the SPL applies to iterates on exactly the same footing as to primitive geodesics, since the shift length $m\ell_\gamma$ enters the SPL as a scalar parameter $r = m\ell_\gamma/L$. By Remark 4.3, iterates with $m\ell_\gamma > 2L$ contribute exactly zero and need not be handled separately. Hence each individual geodesic summand

(primitive or iterated), weighted by the positive coefficient $\ell_\gamma/(2 \sinh(m\ell_\gamma/2))$, contributes negatively to the even-minus-odd difference.

Step 2 (frontier dominance). By (14) and (16), the aggregate contribution of frontier geodesics to $\Delta^\Gamma(\lambda)$ is negative of order $-c_\Gamma\sqrt{\lambda}$, while the combined contribution of non-frontier geodesics plus the archimedean term is $O(\log^2 \lambda)$ (following the same arguments as in Part II of the Riemann case, adapted for the hyperbolic Plancherel kernel).

Step 3 (simplicity). Simplicity of the minimum eigenvalue of QW_λ^Γ follows either from

- the Jentzsch–Perron argument applied to the positivity of the regularized hyperbolic geodesic kernel (which is positive, since $\ell/(2 \sinh(\ell/2)) > 0$ for all $\ell > 0$), or
- directly from Selberg’s spectral theory: the minimum of QW_λ^Γ corresponds spectrally to the lowest non-constant Laplace eigenvalue $\lambda_1 > 0$, which is simple for generic Γ and in any case yields a simple minimum of the associated quadratic form up to numerical degeneracy.

The two arguments agree; here the redundancy begins to show.

Step 4 (Connes’ reduction). Theorem 2.1 is the *algebraic mechanism*: given ingredients (i) a lower-bounded self-adjoint operator on a symmetric interval and (ii) a simple minimum with even eigenfunction, it forces all zeros of the minimum eigenvector’s Fourier transform onto the real line. Verifying ingredients (i) and (ii) for QW_λ^Γ is the $C2^\Gamma$ substep. For Selberg, *both* are trivialised by the Lie-group structure: (i) follows from the compactness of M plus standard Laplacian theory, and (ii) follows from positivity of the hyperbolic geodesic kernel and the spectral theorem applied to Δ_Γ . (In the three-component decomposition $\text{RH}_X = \text{GBC}_X + C2_X + \text{Hurwitz}_X$ developed in the companion work [12], in preparation, these correspond to the Lie-origin branch.) For Riemann, ingredient (ii) is non-trivial; it is the content of v2.1 Proposition A6 [11]. Applying Connes reduction with QW_λ^Γ in place of QW_λ , the zeros of the Selberg zeta function are real (on the critical line under the substitution $s = 1/2 + ir$). This is Selberg-RH. $\square$

Remark 4.11 (Conditionality and consistency). Theorem 4.10 is a conditional reduction: *given* even dominance, Selberg-RH follows by the v2.0 argument. Since Selberg-RH is already known unconditionally via Theorem 3.2, the classical proof simultaneously verifies the conclusion. The value of the conditional reduction lies in *certifying the universality of the v2.0 method*: it applies in a case where a Hilbert–Pólya operator already exists, and the two routes reach the same conclusion by independent means. This agreement is evidence of consistency.

Remark 4.12 (Even dominance as structural expectation). The even-dominance assumption in Theorem 4.10 is a *structural expectation*: it follows from the positivity of the hyperbolic geodesic kernel $\ell/(2 \sinh(\ell/2)) > 0$ and the Jentzsch–Perron argument in Step 3 of the proof sketch above. A fully rigorous proof would require the same frontier-dominance estimates as in the Riemann Part II [11], adapted to the exponential geodesic density. Lemma 4.7 provides the key quantitative input. For Selberg, the structural expectation is additionally supported by the independent operator-theoretic proof (Theorem 3.2), which confirms the conclusion without using even dominance at all.

4.5 NE-A for Selberg

Define the Selberg prime shift multiplier

$$M^\Gamma(\xi) = -2 \operatorname{Re} \left[\frac{Z'_\Gamma}{Z_\Gamma} \left(\frac{1}{2} + i\xi \right) \right]. \quad (18)$$

Proposition 4.13 (NE-A for Selberg). *The set $\{\xi \in \mathbb{R} : M^\Gamma(\xi) < 0\}$ has positive Lebesgue measure.*

Proof sketch. By the Hadamard product representation of the Selberg zeta function on a compact hyperbolic surface ([15, Ch. 10]), one has

$$\frac{Z'_\Gamma}{Z_\Gamma}(s) = b_\Gamma + \sum_{\rho} \left[\frac{1}{s - \rho} + \frac{1}{\rho} \right] + (\text{contributions from trivial zeros and the topological factor}), \quad (19)$$

where ρ runs over the non-trivial zeros (on $\operatorname{Re}(s) = 1/2$ under Theorem 3.2) and $b_\Gamma \in \mathbb{R}$ is a constant depending on Γ .

Setting $s = 1/2 + i\xi$ and taking real parts, each non-trivial zero $\rho_j = 1/2 + ir_j$ contributes

$$\operatorname{Re} \left[\frac{1}{i(\xi - r_j)} + \frac{1}{1/2 + ir_j} \right] = \frac{1/2}{1/4 + r_j^2} > 0. \quad (20)$$

Thus the non-trivial zeros produce a *positive constant* contribution to $\operatorname{Re}[Z'_\Gamma/Z_\Gamma(1/2 + i\xi)]$; they do not create sign oscillations in $M^\Gamma(\xi)$.

The sign oscillations arise from the trivial zeros (at $s = -k$, $k \geq 0$) and the topological factor, whose contributions to $\operatorname{Re}[Z'_\Gamma/Z_\Gamma]$ are of the form $\sum_{k \geq 0} c_k / [(\xi^2 + (k + 1/2)^2)]$ with $c_k \in \mathbb{R}$ of alternating sign. Combining with the explicit-formula identity term of the Selberg trace formula (7), one finds by a direct application with a suitable test function (localising in ξ) that $M^\Gamma(\xi)$ takes both strictly positive and strictly negative values on sets of positive Lebesgue measure. The argument is the direct analogue of [11, Lem. A.3]; the only substitution is $\tanh(\pi r)$ in place of $\Gamma'/\Gamma(1/4 + ir/2)$ in the archimedean kernel. $\square$

NE-A therefore transfers. The prime shift multiplier is oscillatory in both settings. This is a structural, generic feature of any explicit-formula zeta function: the zeros contribute sign-indefinite oscillations to the logarithmic derivative, regardless of whether those zeros are of arithmetic or geometric origin.

5 NE-B Fails for Selberg: The Universal Commutant Δ_Γ

Here we come to the substantive difference between Selberg and Riemann.

5.1 Statement

Theorem 5.1 (NE-B failure for Selberg). *Let $M = \Gamma \backslash \mathbb{H}$ be a compact hyperbolic surface, and denote by Δ_Γ the Laplace–Beltrami operator. Then for every $\gamma \in \mathcal{P}_\Gamma$ and every $m \geq 1$, the geodesic shift operator $T_{m\ell_\gamma}^\Gamma$ (defined below) commutes with Δ_Γ :*

$$[\Delta_\Gamma, T_{m\ell_\gamma}^\Gamma] = 0. \quad (21)$$

Consequently the family $\{T_{m\ell_\gamma}^\Gamma\}_{\gamma, m}$ admits a universal non-trivial commutant. The Selberg analogue of NE-B (Theorem 2.6) is false.

5.2 The geodesic shift operator on $L^2(M)$

The objects that v2.0 manipulates are shifts on the *logarithmic variable* $t \in [-L, L]$. In the Riemann setting, the variable t corresponds to $\log x$ on the idele class space (in Connes' adelic framework). In the Selberg setting, the analogous variable is directly the hyperbolic distance on the surface, and the shifts are implemented as follows.

Definition 5.2 (Geodesic shift operators). Let $\phi^t : SM \rightarrow SM$ denote the geodesic flow on the unit tangent bundle SM , with parameter $t \in \mathbb{R}$. For a primitive closed geodesic γ with length ℓ_γ , the m -fold iterate $\phi^{m\ell_\gamma}$ is a translation along geodesic orbits.

On the subspace of $L^2(SM)$ invariant under the right-regular $\mathrm{SO}(2)$ -action (spherical functions on M), the pullback $T_{m\ell_\gamma}^\Gamma := (\phi^{m\ell_\gamma})^*$ is an operator commuting with the Casimir / Laplacian.

More invariantly, the geodesic shift operators are elements of the convolution algebra $C_c^\infty(\Gamma \backslash \mathrm{PSL}(2, \mathbb{R}) / \mathrm{SO}(2))$; this is the bi- $\mathrm{SO}(2)$ -invariant convolution algebra acting on $L^2(M)$ via the regular representation. It is commutative (a classical consequence of Gelfand's theorem on bi- K -invariant functions on a semisimple group), and the Laplacian Δ_Γ is in its commutant as a differential operator commuting with all left-translations.

5.3 Proof of Theorem 5.1

Proof. We prove (21) by identifying both Δ_Γ and the geodesic shift operators within the algebra of left-invariant operators on $\Gamma \backslash \mathrm{PSL}(2, \mathbb{R})$ acting on the spherical subspace.

Step 1 (invariance under isometries). The Laplacian Δ_Γ is invariant under the full isometry group of M , which contains the subgroup of *geodesic translations* (flows along the geodesic direction). This is the defining property of the Laplace–Beltrami operator: it is the unique (up to scalar) second-order differential operator invariant under all isometries.

Step 2 (geodesic shift as left-translation). The geodesic flow ϕ^t on SM corresponds to the right-action of the one-parameter subgroup

$$A = \left\{ \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} : t \in \mathbb{R} \right\} \subset \mathrm{PSL}(2, \mathbb{R}). \quad (22)$$

Since Δ_Γ corresponds to the Casimir element of the universal enveloping algebra $\mathcal{U}(\mathfrak{sl}_2)$, which is in the center of $\mathcal{U}$, it commutes with every element of the regular representation of $\mathrm{PSL}(2, \mathbb{R})$ — in particular with the one-parameter subgroup A .

Step 3 (closed-geodesic translates). Primitive closed geodesics of length ℓ_γ correspond to hyperbolic conjugacy classes $[h_\gamma] \subset \Gamma$. The translation along γ by parameter $m\ell_\gamma$ sends a point on γ back to itself (closure). The operator $T_{m\ell_\gamma}^\Gamma$ is the restriction of the A -action by $t = m\ell_\gamma$ to the spherical subspace of $L^2(\Gamma \backslash \mathrm{PSL}(2, \mathbb{R}))$.

Step 4 (commutation). Combining Steps 2 and 3: since Δ_Γ is the Casimir and thus central, and $T_{m\ell_\gamma}^\Gamma$ is a right-regular action of A , they commute as operators on $L^2(M)$:

$$[\Delta_\Gamma, T_{m\ell_\gamma}^\Gamma] = 0.$$

This holds for every γ and every $m \geq 1$. Hence Δ_Γ is a non-trivial universal commutant. The Selberg analogue of NE-B fails. $\square$

Remark 5.3 (Alternative elementary argument). A more direct argument: decompose $L^2(M)$ in the eigenbasis $\{\psi_j\}$ of Δ_Γ with eigenvalues $\lambda_j = 1/4 + r_j^2$. The action of any geodesic flow element ϕ^t on spherical functions is diagonal in this basis, with eigenvalues e^{itr_j} (this is the essential content of the Selberg transform). In particular $T_{m\ell_\gamma}^\Gamma \psi_j = e^{im\ell_\gamma r_j} \psi_j$. Both Δ_Γ and $T_{m\ell_\gamma}^\Gamma$ act diagonally in the common eigenbasis, so they commute.

Remark 5.4 (Why this is possible for Selberg but not for Riemann). The argument depends on the existence of a *common geometric space* (the surface M) on which all geodesics live, together with a *continuous symmetry group* (the isometry group of $\mathbb{H}$, or equivalently $\mathrm{PSL}(2, \mathbb{R})$) that acts transitively on unit tangent vectors. Primes, by contrast, do not live on a common geometric space — they are arithmetic objects without a continuous symmetry that permutes them. This is what NE-B captures: multiplicative independence implies absence of hidden common symmetry, which implies absence of a universal commutant.

5.4 Consequence for the numerical commutator test

In the Selberg case, repeating the numerical commutator test of [11, Section 5.3] would yield a *large* kernel for the commutator constraint matrix C_N — in fact a kernel whose dimension grows with the number of distinct Laplace eigenvalues sampled, because every polynomial in Δ_Γ is a commutant. The singular-value spectrum would show no “isolated-1-dim-null”-structure; instead many small singular values would appear, reflecting the infinite-dimensional commutant.

This is the diagnostic: *if you run the v2.0 commutator test on a genuine Hilbert–Pólya system, it detects the operator*. In the Riemann case, the test detects nothing nontrivial — and this is, by the logic of v2.0, precisely because no such operator exists.

6 Hilbert–Pólya as a Special Case of v2.0

We now articulate the meta-principle that emerges from combining the Riemann and Selberg analyses.

6.1 The Hilbert–Pólya programme revisited

The classical Hilbert–Pólya conjecture proposes to prove RH by constructing a self-adjoint operator H on a Hilbert space whose spectrum is the set of imaginary parts $\{\gamma_k\}$ of the non-trivial Riemann zeros. Such an operator, if it exists, would give RH as an immediate corollary (self-adjointness $\Rightarrow$ real eigenvalues $\Rightarrow$ zeros on the critical line).

One hundred years of effort have produced candidate operators (Berry–Keating $H = xp$ [5], Connes’ adelic Dirac operator [7], and others) but none that provably realize the Riemann spectrum.

6.2 The v2.0 alternative

The v2.0 programme circumvents the Hilbert–Pólya requirement by establishing RH through a *quadratic-form inequality* (even dominance of QW_λ), rather than through an operator-spectral identification. The ingredients are:

- A parity decomposition of the ambient Hilbert space.

- An algebraic *pointwise* parity advantage (Shift Parity Lemma).
- An analytic *aggregation* mechanism (frontier dominance).
- An argument that no simpler route applies (NE-A, NE-B).

Crucially, *no operator is constructed*. The route is via the quadratic form directly.

6.3 Selberg as validation: the operator exists but is not required

Theorem 4.10 shows that v2.0 conditionally reduces Selberg-RH to even dominance of QW_λ^Γ . In this case, a Hilbert–Pólya operator Δ_Γ also exists (Theorem 5.1), and its spectral positivity independently confirms the conclusion. The two routes coexist:

- Selberg’s original proof (spectral: $\Delta_\Gamma \geq 0$, unconditional).
- The v2.0 reduction (quadratic: even dominance of QW_λ^Γ implies Selberg-RH, conditional on even dominance).

Both arrive at the same conclusion by different routes. The v2.0 method *subsumes* the operator-based route: it applies whether or not a Hilbert–Pólya operator exists, and where one does, it yields the same result.

6.4 Riemann as the complementary case

In the Riemann setting, NE-B (in the full-space conjectural form) asserts that no universal commuting operator exists. If this conjecture holds, then Hilbert–Pólya cannot be realized in the classical sense, and yet RH holds by v2.0. This is the *scientifically interesting* case: v2.0 provides a proof route that does not demand the operator’s existence.

6.5 Meta-theorem

Theorem 6.1 (v2.0 meta-principle, informal). *Let Z be a zeta-like function admitting an explicit formula of the Riemann–Weil type (with real-valued logarithmic shifts and positive weights). Then:*

- (i) *The v2.0 ingredients (Shift Parity Lemma, frontier dominance, parity decomposition of the associated Weil quadratic form) transfer to Z with modifications only in the density and weight constants.*
- (ii) *The v2.0 route to the corresponding Riemann Hypothesis is available whenever an appropriate frontier-density bound holds.*
- (iii) *A Hilbert–Pólya operator for Z exists if and only if the analogue of NE-B fails, i.e. the shift operators associated to Z admit a non-trivial universal commutant.*
- (iv) *When such an operator exists, v2.0 is redundant (but valid). When it does not, v2.0 is the only known proof route.*

We formulate this as a *structural principle* rather than a formal theorem, because it depends on what one takes as the admissible class of zeta-like functions. For classical L-functions satisfying a standard functional equation and explicit formula (Selberg class, Connes’ family), all four items are expected to hold under the stated conditions.

Remark 6.2 (Refinement in the companion meta-paper). The informal meta-principle of Theorem 6.1 has been made technically precise in the companion paper [12] as the *Meta-Theorem* (Thm. thm:meta there), via the three-component decomposition

$$\text{RH}_X = \text{GBC}_X + C2_X + \text{Hurwitz}_X.$$

In that language, items (iii)–(iv) of our informal Theorem 6.1 are the statement that $C2_X$ is trivialised in the Lie-origin branch and open (equivalent to classical Hilbert–Pólya) in the arithmetic branch. The finer point noted in [12] is that $C2_X$ itself decomposes into a *parity substep* $C2_X^{\text{parity}}$ (simple ground state with even eigenvector) and an *eigenvector-realisation substep* $C2_X^{\text{eigenvector}}$ (Hermite-prolate approximation convergence). In the current CORE status, the Riemann $C2/MS2$ block is closed in the CCM/Fourier branch by the Zookeeper three-lemma microcluster closure. This does not change the Selberg proof obligation: for Selberg, both $C2$ substeps are trivialised simultaneously by the Laplace–Casimir. The Selberg case is therefore a positive control for the normal form, not a dependent corollary of the Riemann closure.

6.6 Restatement: the Weil quadratic form is fundamental

The structural insight is:

The Weil quadratic form QW is the fundamental object; the Hilbert–Pólya operator is a realization that may or may not exist.

For one hundred years, the Riemann Hypothesis has been viewed through the Hilbert–Pólya lens: find the operator, prove RH. The v2.0 framework inverts this view: prove even dominance, prove RH, and investigate (as a separate question) whether a Hilbert–Pólya operator happens to exist. In the Selberg case it does; in the Riemann case (by NE-B) it does not. Both cases are captured by the same underlying framework.

7 Applications and Outlook

7.1 Generalized L-functions

The v2.0 method applies in principle to any zeta-like function with a Riemann–Weil-type explicit formula. We expect the following:

- *Dirichlet L-functions* $L(s, \chi)$ for a Dirichlet character χ : the explicit formula is the prime summation weighted by $\chi(p)$. The Shift Parity Lemma transfers (the character introduces a phase, not a sign-change of the geometric parity structure). Frontier dominance transfers (with PNT replaced by the PNT for L -functions, Landau–Siegel bounds). NE-B: we conjecture it continues to hold, since χ does not introduce a new hidden symmetry. In the current Zeta-Zoo cartography, Dirichlet is not closed merely by importing the untwisted Riemann package: it remains the Weg-D / χ -twisted CCM follow-up branch. A concrete Dirichlet write-up would make the meta-framework’s predictions operationally testable — it is the natural next step after the present Selberg consistency check.
- *Dedekind zeta functions* $\zeta_K(s)$ for a number field K : the explicit formula sums over prime ideals $\mathfrak{p}$ with weight $(\log N\mathfrak{p}) \cdot N\mathfrak{p}^{-m/2}$. The Shift Parity and frontier mechanisms transfer. NE-B: conjecturally holds for fields that are not CM-multiplication (where an extra symmetry could interfere).

- *Automorphic L-functions* (Hecke, $\mathrm{GL}(n)$ -cuspidal): the explicit formula transfers. A Hilbert–Pólya operator exists precisely when the L-function has a spectral interpretation via a Laplacian (which happens when the automorphic form is itself of geometric origin — e.g. Maass forms on a hyperbolic surface). Otherwise NE-B-analogue is expected.

7.2 Ruelle zeta for Axiom-A flows

For an Axiom-A flow on a compact Riemannian manifold (a hyperbolic dynamical system), the Ruelle zeta function is defined by

$$\zeta_R(s) = \prod_{\tau} \frac{1}{1 - e^{-s\ell_{\tau}}}, \quad (23)$$

where the product is over primitive periodic orbits τ with minimal period ℓ_{τ} . A Ruelle–Riemann hypothesis (analogue of the Selberg case) has been proven for smooth Anosov flows [10]. For non-hyperbolic flows, the analogue is conjectural.

Conjecture 7.1 (v2.0 Ruelle). *For a hyperbolic flow with topological entropy h , the v2.0 method applies to the Ruelle Weil quadratic form QW_{λ}^R and yields a proof of the critical-line statement $\mathrm{Re}(s) = h/2$ for all non-trivial zeros of ζ_R . NE-B holds in this generic setting (no universal Laplacian), so the proof is essentially unique.*

The Conjecture is sharper than Selberg-RH in that it does not rely on an underlying Riemannian symmetric structure. For generic Axiom-A flows, no Laplacian is available; the v2.0 method would be the only route.

7.3 Ihara zeta and the discrete–continuum axis

The Ihara zeta function $\zeta_{\Gamma}(s)$ of a finite connected $(q+1)$ -regular graph Γ is defined by

$$\zeta_{\Gamma}(s)^{-1} = (1 - s^2)^{r-1} \det(I - sA + qs^2I), \quad (24)$$

where A is the adjacency matrix and r is the cycle rank of Γ [19, 20, 21]. The determinant formula is sometimes called the Ihara–Bass formula; the proofs by Hashimoto (1989) and Bass (1992) established it in full generality. The Ihara RH is the statement that all non-trivial poles lie on the circle $|s| = q^{-1/2}$, equivalent to Γ being a Ramanujan graph. It has been proved for explicit Ramanujan families constructed from $\mathrm{PGL}(2)$ -arithmetic [22].

In the v2.0 / Zookeeper framework, the Ihara case is *discrete–operatorial* (SGE-YES), in parallel with the *continuous–operatorial* Selberg case. The graph Laplacian $L = (q+1)I - A$ commutes with all automorphisms of Γ and hence with the edge-adjacency transfer matrix T . This makes L the discrete analogue of the hyperbolic Laplacian Δ_{Γ} : a universal commutant that renders NE-B vacuous for Ihara, for exactly the same structural reason as in the Selberg case.

The three-way comparison axis in the Zookeeper framework is:

Zeta function	Type	Commutant	v2.0 class
Selberg $Z_{\Gamma}(s)$	continuous, geometric	Δ_{Γ} (Laplace–Beltrami)	SGE-YES
Ihara $\zeta_{\Gamma}(s)$	discrete, combinatorial	$L = (q+1)I - A$ (graph Laplacian)	SGE-YES
Riemann $\zeta(s)$	arithmetic	none (NE-B holds)	SGE-NO

In both SGE-YES cases the zeta-function’s RH is a consequence of a classical spectral bound (Selberg’s $\lambda_1 > 1/4$; Ramanujan’s $|\text{eigenvalue}| \leq 2\sqrt{q}$), and the v2.0 framework provides a *consistent re-derivation* via even dominance of the corresponding Weil quadratic form. For the Riemann zeta, no such commutant exists; the v2.0 method is the only available structural route to RH.

7.4 Physical interpretation

The Selberg setting corresponds physically to a quantum particle on a hyperbolic surface (“quantum chaos”); the Laplace eigenvalues are the energy levels. The v2.0 Shift Parity Lemma states that the parity of the eigenfunctions is biased toward even functions at the threshold. This is a statement about the parity-class distribution of quantum eigenstates.

For Axiom-A flows (billiards, chaotic cavities), the v2.0 framework predicts GUE-type statistics for the Ruelle zeros, as in the BGS (Bohigas–Giannoni–Schmit) conjecture [6]; see also [3] for related GUE statistics of zeros of principal L -functions. The v2.0 methodology may offer a structural explanation of this conjecture.

8 Zookeeper Consistency: Selberg as an SGE-YES Test

The Selberg case is also a useful calibration point for the post-Zookeeper version of the Zeta Zoo. In this language, Selberg is an SGE-YES system: the relevant operator exists classically, and the v2.0 transfer must therefore reproduce a known positive case rather than create a new assertion from scratch.

Table 1 records the five transfer slots and their occupants in the Selberg and Riemann settings.

Table 1: The five SGE-YES transfer slots for Selberg vs. Riemann. A tick (✓) indicates the slot is classically occupied; “v2.0” indicates it is established by the v2.0 framework.

Slot	Selberg (SGE-YES)	Riemann
Operator / form	Δ_Γ (classical; commutes with all geodesic shifts; NE-B <i>fails</i>) ✓	QW_λ (no classical HP operator; NE-B holds)
Defect	Discrepancy between QW_λ^Γ and Δ_Γ -spectral positivity ✓	Gap to even dominance (conjectural $\Rightarrow$ established by v2.0)
Approximation	Length cutoffs, truncated trace formula ✓	Galerkin truncation $D_N(r)$ (v2.0)
Gap	Spectral gap $\lambda_1(\Delta_\Gamma) > \frac{1}{4}$ ✓	Even-dominance gap $\Delta(\lambda) > 0$ (v2.0)
Limit	Full Selberg trace formula [17] ✓	Full Weil explicit formula

Remark 8.1 (Galerkin vs. spectral truncation). In the Riemann case the approximation is a *Galerkin cutoff*: the infinite-dimensional quadratic form QW_λ is approximated by the finite matrix $D_N(r)$ via a cosine/sine basis of size N . For Selberg the approximation is

instead a *spectral cutoff*: the geodesic sum is truncated at length $\ell_\gamma \leq 2L$, and Remark 4.3 shows this truncation is exact, not approximate. The Zookeeper *Approximation* slot therefore uses different but analogous technology in the two cases.

Thus Selberg plays a diagnostic role. If the Zookeeper normal form did not detect a positive gap here, the mapping would be wrong. Conversely, the fact that the Laplacian commutes with the geodesic shifts explains why NE-B fails in the Selberg setting and highlights the contrast with the Riemann case: for Selberg the Hilbert–Polya operator is present; for Riemann, the original Hilbert–Polya commutant route is structurally blocked while the separate Zookeeper package closes the Riemann *C2/MS2* kernel in the CCM/Fourier branch. Selberg therefore validates the SGE-YES side of the boundary theorem; it is not an additional open blocker and not evidence that the Prime-Hub/OP5 graph problem has been solved.

Proposition 8.2 (Selberg–Zookeeper Consistency). *The v2.0 framework correctly classifies the Selberg zeta function $Z_\Gamma(s)$ as an SGE-YES system. Specifically:*

- (i) *Even dominance of QW_λ^Γ implies the Selberg Riemann Hypothesis (Theorem 4.10).*
- (ii) *The Laplace–Beltrami operator Δ_Γ is a universal commutant, so NE-B fails for Selberg (Theorem 5.1).*
- (iii) *All five Zookeeper transfer slots are occupied (Table 1).*
- (iv) *The v2.0 classification is consistent with and independently verifies Selberg’s classical spectral proof via Δ_Γ .*

9 Conclusion

We have tested the v2.0 methodology on the Selberg zeta function. The Shift Parity Lemma, frontier dominance, the Weil quadratic form, and NE-A all transfer to the Selberg setting with modest modifications (mostly constants and density regimes). Even dominance of the Selberg Weil quadratic form QW_λ^Γ yields a second proof of the Selberg Riemann Hypothesis, redundant in view of Selberg’s spectral proof.

The decisive structural finding is that *NE-B fails for Selberg*: the Laplace–Beltrami operator Δ_Γ is a universal commutant for the family of geodesic shift operators. This is the mark of a genuine Hilbert–Pólya system — and the v2.0 commutator test, run on such a system, would detect it.

Combining the Riemann and Selberg pictures, we formulate a meta-principle: the v2.0 framework is *universal*, and the Hilbert–Pólya programme is a *special case* of it, realized precisely when the analogue of NE-B fails. In the Selberg case it fails, and the operator is Δ_Γ . In the Riemann case it holds (at least for the truncations tested and conjecturally in general), and no Hilbert–Pólya operator exists; yet v2.0 still yields RH, through the quadratic form route alone.

Principle. *The Weil quadratic form is the fundamental object. The operator is optional.*

A century of thought has identified the operator as the heart of Riemann’s conjecture. The v2.0 framework suggests instead that the quadratic-form inequality is the heart, and the operator, where it exists, is its geometric by-product.

References

- [1] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. (N.S.) **1** (1995), 411–457. doi:[10.1007/BF01589495](https://doi.org/10.1007/BF01589495).
- [2] C. Deninger, *Motivic L-functions and regularized determinants*, in *Motives* (Seattle, WA, 1991), Proc. Sympos. Pure Math. **55**, Part 1, Amer. Math. Soc., Providence, RI, 1994, pp. 707–743.
- [3] Z. Rudnick and P. Sarnak, *Zeros of principal L-functions and random matrix theory*, Duke Math. J. **81** (1996), no. 2, 269–322. doi:[10.1215/S0012-7094-96-08115-6](https://doi.org/10.1215/S0012-7094-96-08115-6).
- [4] A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, Comm. Sémin. Math. Univ. Lund (dédié à M. Riesz) (1952), 252–265.
- [5] M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Review **41** (1999), 236–266. doi:[10.1137/S0036144598347497](https://doi.org/10.1137/S0036144598347497).
- [6] O. Bohigas, M.-J. Giannoni, and C. Schmit, *Characterization of chaotic quantum spectra and universality of level fluctuation laws*, Phys. Rev. Lett. **52** (1984), 1–4.
- [7] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.
- [8] A. Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022 (February 2026). [arXiv:2602.04022](https://arxiv.org/abs/2602.04022).
- [9] A. Connes and H. Moscovici, *Prolate spheroidal operator and Zeta*, Proc. Natl. Acad. Sci. USA **119** (2022), no. 28, e2123174119. doi:[10.1073/pnas.2123174119](https://doi.org/10.1073/pnas.2123174119). Also: arXiv:2112.05500.
- [10] D. Dolgopyat, *On decay of correlations in Anosov flows*, Ann. of Math. (2) **147** (1998), 357–390.
- [11] L. Geiger, *The Riemann Hypothesis: Even Dominance of the Weil Quadratic Form*, Zenodo preprint (2026). DOI: 10.5281/zenodo.19546593 (v2.1).
- [12] L. Geiger, *The Meta-Galerkin Bridge: Explicit Formula Principle and the Three-Component Decomposition of Riemann-Hypothesis-Type Statements*, companion paper (2026, in preparation).
- [13] D. A. Hejhal, *The Selberg Trace Formula for $\mathrm{PSL}(2, \mathbb{R})$* , Vol. 1, Lecture Notes in Mathematics **548**, Springer-Verlag, 1976.
- [14] H. Huber, *Zur analytischen Theorie hyperbolischer Raumformen und Bewegungsgruppen*, Math. Ann. **138** (1959), 1–26.
- [15] H. Iwaniec, *Spectral Methods of Automorphic Forms*, 2nd ed., Graduate Studies in Mathematics **53**, Amer. Math. Soc., 2002.
- [16] P. Sarnak, *Asymptotic behavior of periodic orbits of the horocycle flow and Eisenstein series*, Comm. Pure Appl. Math. **34** (1981), 719–739.

- [17] A. Selberg, *Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series*, J. Indian Math. Soc. **20** (1956), 47–87.
- [18] A. Selberg, *On the estimation of Fourier coefficients of modular forms*, Proc. Sympos. Pure Math. **VIII**, Amer. Math. Soc., Providence, RI, 1965, pp. 1–15.
- [19] Y. Ihara, *On discrete subgroups of the two by two projective linear group over p -adic fields*, J. Math. Soc. Japan **18** (1966), no. 2, 219–235.
- [20] K. Hashimoto, *Zeta functions of finite graphs and representations of p -adic groups*, in *Automorphic Forms and Geometry of Arithmetic Varieties*, Adv. Stud. Pure Math. **15**, Academic Press, 1989, pp. 211–280.
- [21] H. Bass, *The Ihara–Selberg zeta function of a tree lattice*, Int. J. Math. **3** (1992), no. 6, 717–797.
- [22] A. Lubotzky, R. Phillips, and P. Sarnak, *Ramanujan graphs*, Combinatorica **8** (1988), no. 3, 261–277.